

Two-time Nobel Prize-winning theoretical physicist John Bardeen's knowledge of semiconductors was vital to creating the first working transistor. His expertise made the technology usable, launching a revolution in electronics: transistors are integral to every modern electronic device, from radios to space shuttles.

MILESTONES

SEEKS SOLUTION

Realizes the flaw in the original amplifier design; begins work at Bell Labs to find an alternative in 1945.

HAS BREAKTHROUGH

Finishes his "magic month" of experiments in 1947 and produces the point-contact transistor.

TEAM SPLIT

Clashes with Shockley, who stops him from working on interesting projects, and leaves Bell Labs in 1951.

RADIO SHOCK

Hears he has won the Nobel Prize in Physics in 1956 while listening to the radio.

BCS THEORY

Publishes theory of superconductivity with Leon Cooper and John Robert Schrieffer in 1957.

William Shockley (center) hired John Bardeen (right) and Walter Brattain (left) to work on his team at Bell Labs. Bardeen split from the group following patent disputes with Shockley.

John Bardeen grew up in Madison, Wisconsin, and studied electrical engineering. In 1933, at Princeton University, his tutor Eugene Wigner (see p.240) posed the question "How do electrons inside metals interact?" Bardeen made this the subject of his life's work. For his PhD, he calculated a metal's work function (the amount of energy needed to remove an electron). He started to focus more on superconductivity in 1939. It had been known since 1911 that some metals at temperatures approaching absolute zero became superconductors and allowed electricity to pass through them with no resistance. This implied that their electrons were behaving unusually—but Bardeen was not to discover why until the 1950s.

Semiconductors

In 1945, Bardeen joined the solid state group at Bell Telephone Laboratories. The company used glass vacuum tubes to amplify phone signals, but they were bulky and burned out regularly. The group's task was to invent a solid-state amplifier using a semiconductor: a material that acted as an electrical conductor in some conditions but not in others, making it ideal for amplifying an electric current or

